

Synoptic LE: Tree Fluxes 2022

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Table of contents

1	Read in and collate all the files in the Processed Data folder	1
2	Remove bad fluxes and calculate how many there were	1
3	Write out files	4
4	All Data by Year, Site and Zone	4
5	Boxplot summary by Date	6
6	Boxplot summary by Site	8
7	Calculate summary metrics	9
8	Summary Tables	9

1 Read in and collate all the files in the Processed Data folder

2 Remove bad fluxes and calculate how many there were

```
##Remove any samples with Flags in the final qaqc_flag columns
final_dat <- all_dat %>%
  filter(!grepl("TRUE", CH4_rejected_model, ignore.case = TRUE)) %>%
  filter(!grepl("TRUE", CO2_flux_rejected , ignore.case = TRUE))
```

##Removed both CH₄ and CO₂ obs when CO₂ fluxes were negative and when visual checks showed w

```

# count before and after filtering
n_total_co2 <- nrow(all_dat)
n_total_ch4 <- nrow(all_dat)

final_dat <- all_dat %>%
  filter(CH4_rejected_model == FALSE | is.na(CH4_rejected_model)) %>%
  filter(CO2_flux_rejected == FALSE | is.na(CO2_flux_rejected))

n_final <- nrow(final_dat)
n_removed_co2 <- sum(all_dat$CO2_flux_rejected == TRUE, na.rm = TRUE)
n_removed_ch4 <- sum(all_dat$CH4_rejected_model == TRUE, na.rm = TRUE)
n_removed_either <- n_total_co2 - n_final

# build summary table
filter_summary <- tibble::tibble(
  Metric = c(
    "Total observations (joined dataset)",
    "CO2 - observations rejected (n)",
    "CO2 - observations rejected (%)",
    "CH4 - observations rejected (n)",
    "CH4 - observations rejected (%)",
    "Removed by both flags simultaneously (n)",
    "Total unique observations removed (n)",
    "Total unique observations removed (%)",
    "Final observations retained (n)"
  ),
  Value = c(
    n_total_co2,
    n_removed_co2,
    round(100 * n_removed_co2 / n_total_co2, 1),
    n_removed_ch4,
    round(100 * n_removed_ch4 / n_total_co2, 1),
    sum(all_dat$CO2_flux_rejected == TRUE &
      all_dat$CH4_rejected_model == TRUE, na.rm = TRUE),
    n_removed_either,
    round(100 * n_removed_either / n_total_co2, 1),
    n_final
  )
)

filter_summary %>%
  kable(

```

```

caption = "Post-Join QA/QC Filtering Summary",
align   = c("l", "r"),
booktabs = TRUE,
format  = "latex"
) %>%
kable_styling(
  latex_options = c("striped", "hold_position"),
  full_width    = FALSE
) %>%
pack_rows("CO2 flags",          2, 3) %>%
pack_rows("CH4 flags",         4, 5) %>%
pack_rows("Combined filtering", 6, 9)

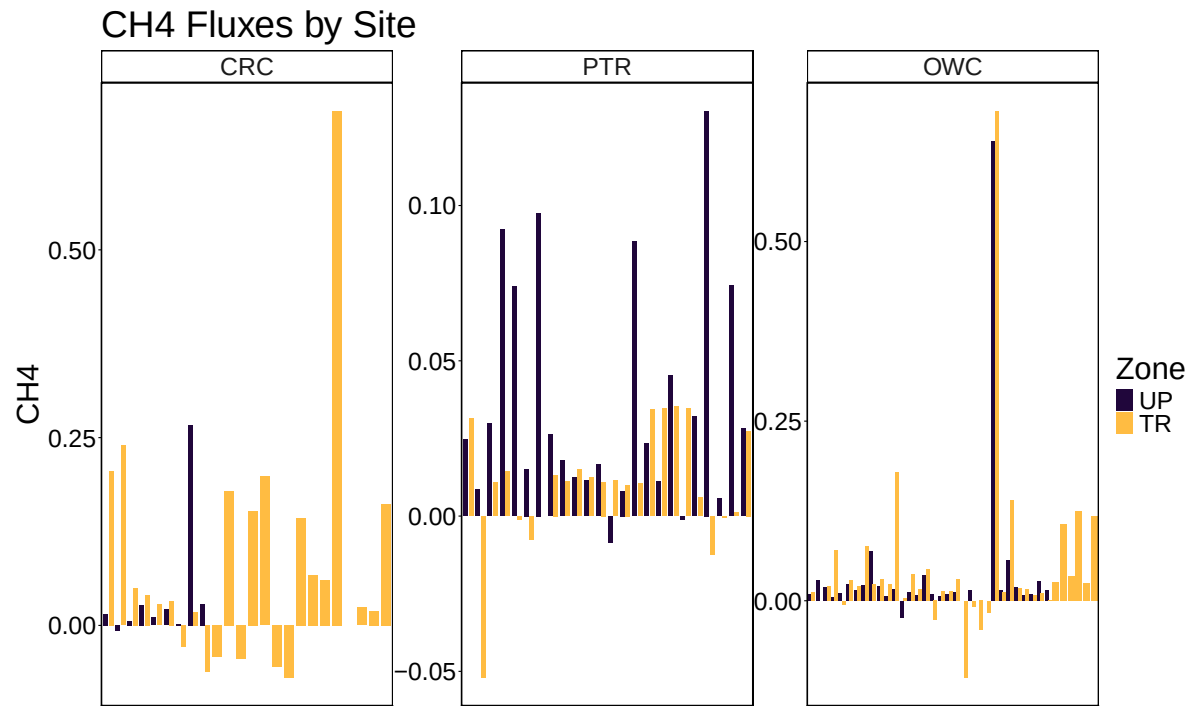
```

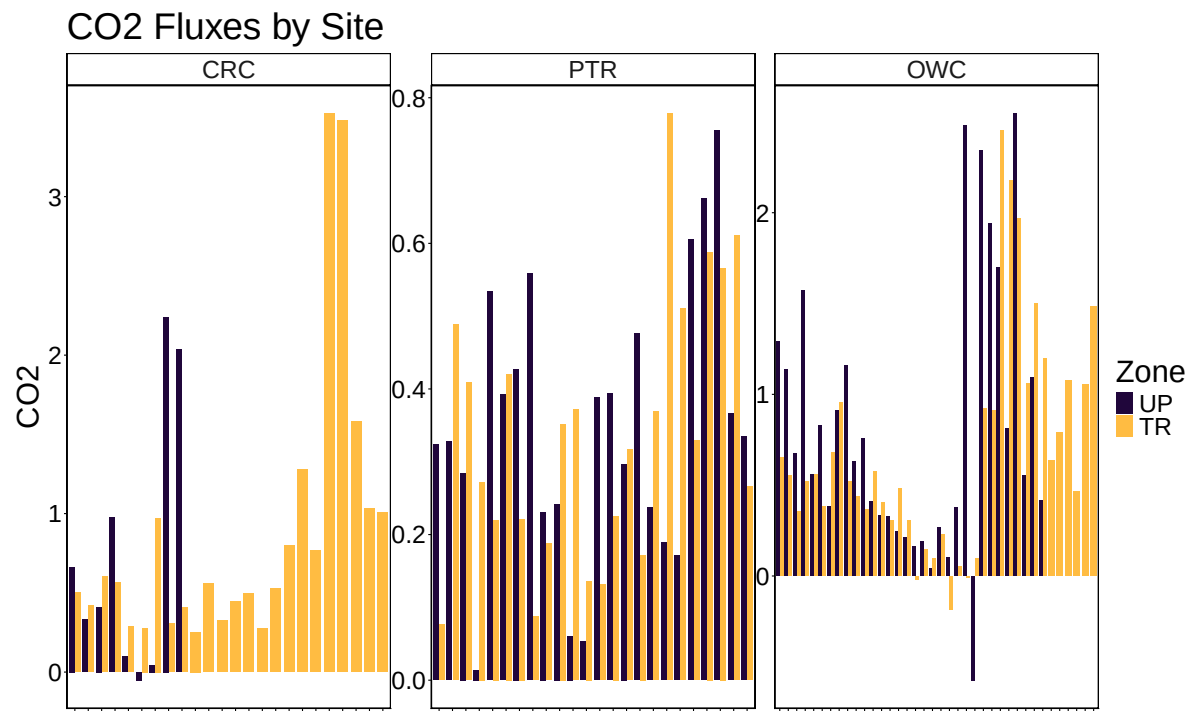
Table 1: Post-Join QA/QC Filtering Summary

Metric	Value
Total observations (joined dataset)	151.0
CO2 flags	
CO2 — observations rejected (n)	8.0
CO2 — observations rejected (%)	5.3
CH4 flags	
CH4 — observations rejected (n)	3.0
CH4 — observations rejected (%)	2.0
Combined filtering	
Removed by both flags simultaneously (n)	1.0
Total unique observations removed (n)	10.0
Total unique observations removed (%)	6.6
Final observations retained (n)	141.0

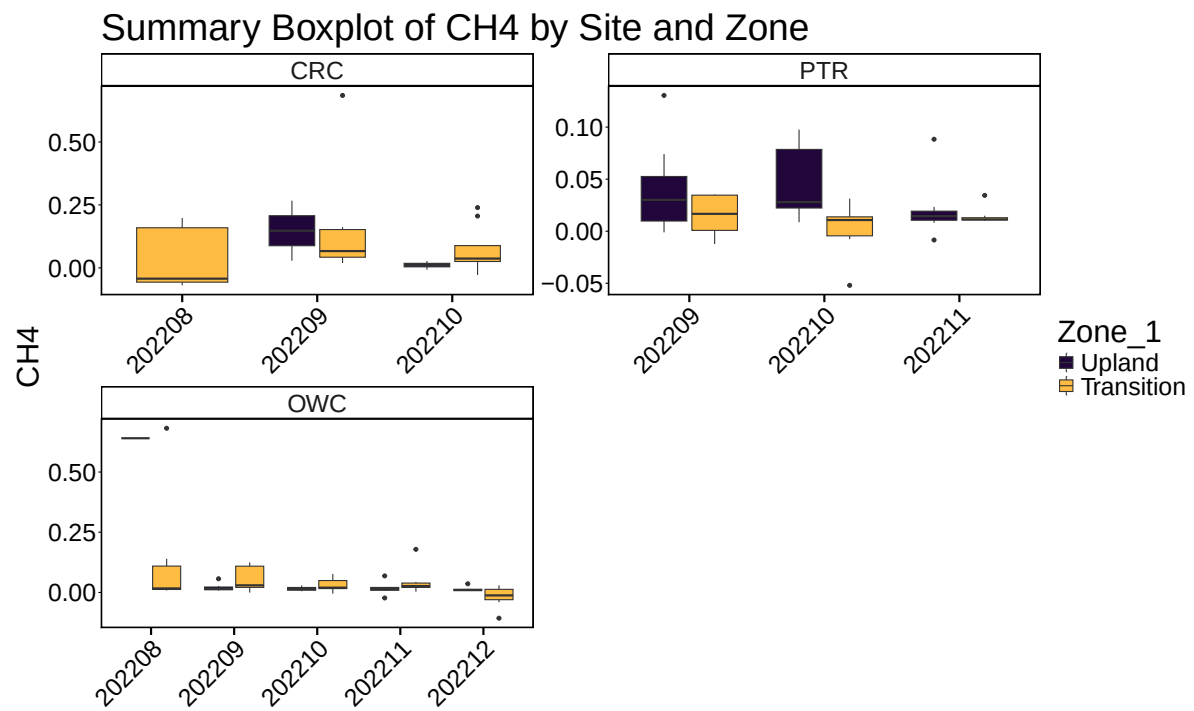
3 Write out files

4 All Data by Year, Site and Zone

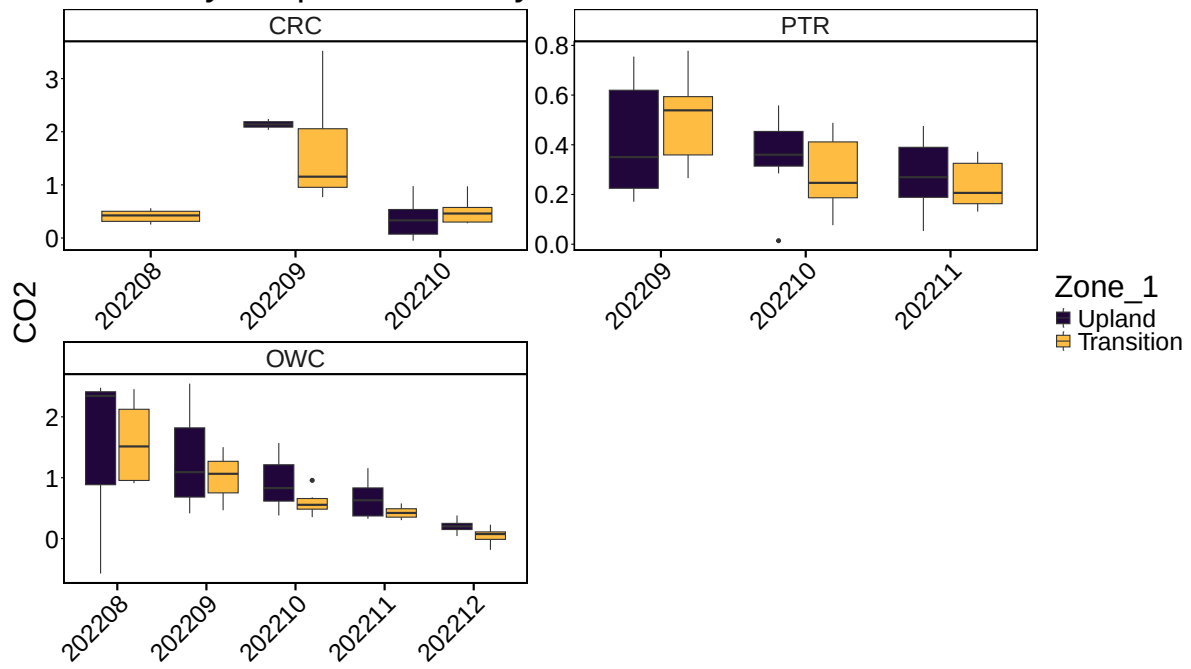




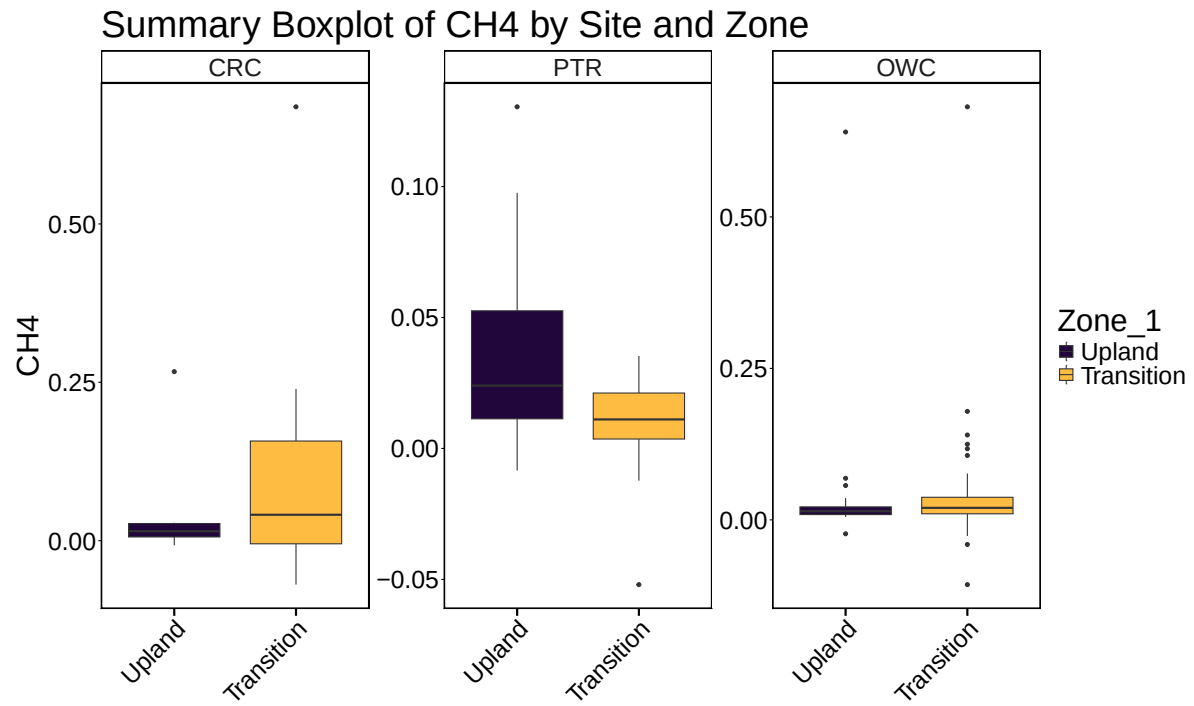
5 Boxplot summary by Date

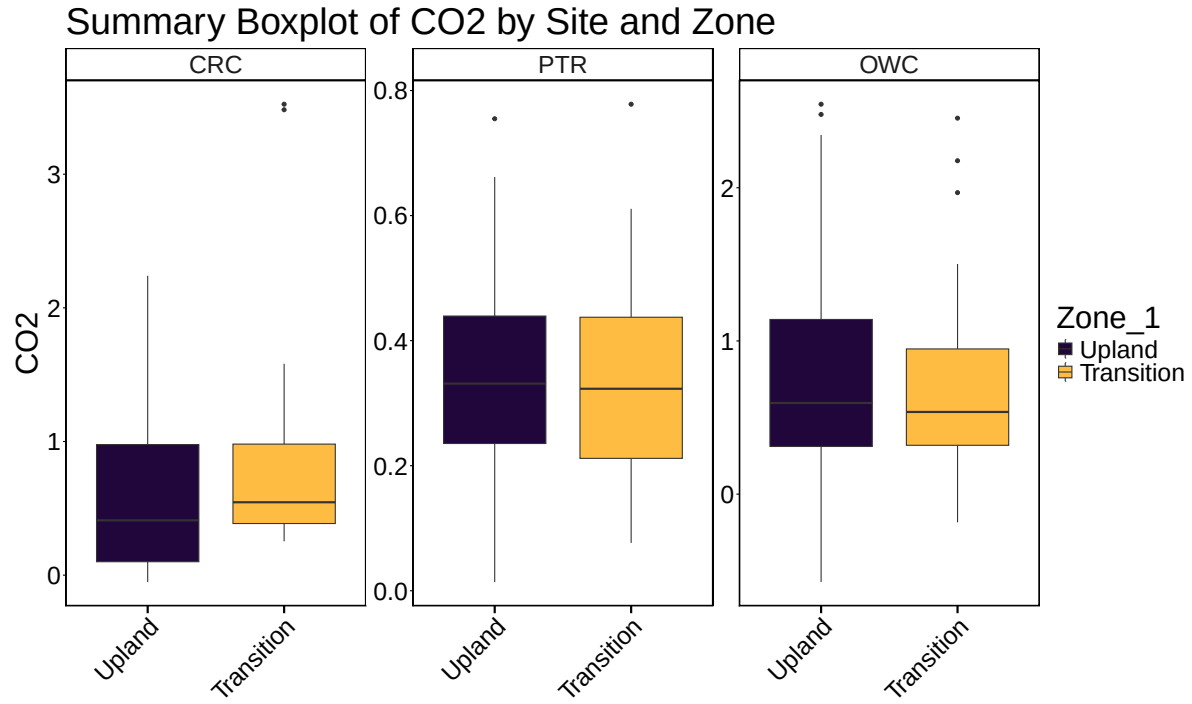


Summary Boxplot of CO2 by Site and Zone



6 Boxplot summary by Site





7 Calculate summary metrics

8 Summary Tables

Table 2: Summary Statistics

Site	Zone_1	min_ch4	max_ch4	mean_ch4	min_co2	max_co2	mean_co2
CRC	Upland	-0.0074088	0.2668133	0.0410601	-0.0511452	2.2403910	0.7503180
CRC	Transition	-0.0693972	0.6848884	0.0871301	0.2534407	3.5239373	0.8622524
PTR	Upland	-0.0084339	0.1304408	0.0359990	0.0140025	0.7548850	0.3468802
PTR	Transition	-0.0519902	0.0352939	0.0109104	0.0764779	0.7781708	0.3377804
OWC	Upland	-0.0229943	0.6401738	0.0383104	-0.5736761	2.5455104	0.8085004
OWC	Transition	-0.1068886	0.6817200	0.0472842	-0.1839348	2.4544127	0.6884196